Ritesh Somashekar

AI Solutions Engineer | Generative AI Services & Adoption

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PROFESSIONAL SUMMARY

- **AI Solutions Engineer** with **5+ years** taking **generative AI** from idea to pilot to supported service in **higher education** and **healthcare**; published at **ACM WSDM '26** and **IEEE PST 2026**.
- Hands-on with **enterprise LLM platforms and APIs (Azure OpenAI, GPT-4, Claude, Gemini)**, **RAG, agents, embeddings, vector search,** and **workflow automation**, built with **security, privacy (HIPAA),** and **access controls (OAuth 2.0)**.
- Partners with researchers, public agencies, and business stakeholders, using **evaluation frameworks**, **documentation**, and **demos** to explain AI capabilities, limitations, and risks to technical and non-technical audiences.

TECHNICAL SKILLS

Generative AI Platforms & APIs: Azure OpenAI, GPT-4, Claude, Gemini, Llama 3 (Ollama), LLM APIs, Prompt Engineering, Structured Outputs

AI Solution Patterns: RAG, Agentic AI, Embeddings, Vector Search (ChromaDB, Pinecone, pgvector, FAISS), LangChain, LlamaIndex, Workflow Automation (Apache Airflow)

AI Evaluation & Governance: Evaluation Frameworks, RAGAS, Hallucination Detection, Guardrails, Model Monitoring, Drift Detection, A/B Testing

Security & Integration: OAuth 2.0, REST APIs, API Versioning, HIPAA Compliance, FastAPI, Docker, Kubernetes, CI/CD

Programming & Cloud: Python, SQL, Java, AWS (S3, Lambda, SageMaker, SQS), Azure, Snowflake, Databricks

Consulting & Delivery: Stakeholder Consultation, Requirements Analysis, Pilot-to-Production Delivery, Technical Documentation, Demonstrations

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, India

PROFESSIONAL EXPERIENCE

George Mason University | ***AI & ML Research Associate*** **Fairfax, VA | August 2024 - Present**

- To bring AI into 9-1-1 training, worked with **Fairfax County DPSC** instructors to scope needs, then co-developed **SHIELD**, an **LLM-agent** training platform; published and demonstrated at **ACM WSDM '26**.
- To move SHIELD from prototype to pilot, supported agency testing sessions and built a **Python/Streamlit** app with **SQLite** accounts and course enrollment, refining it from instructor feedback.
- To deliver safe AI without commercial API costs or data-sharing risk, co-developed **MATE-911**, a locally hosted **RAG** service on **Llama 3.1 8B** with **0.81 protocol F1**; published at **IEEE PST 2026**.
- To make AI output trustworthy in a high-stakes domain, led the **guardrails and validation layer** aligned to **NENA** protocols, lowering hallucinations from **9.5% to 3.0%**.
- To give partners a clear basis for decisions, built an **evaluation framework** with ablation studies, showing **semantic search** retrieval lifted the pass rate from **49.7% to 97.6%**.
- To ground answers in approved content, led the **RAG knowledge base and data pipeline** for **160 gold-standard scenarios**, cutting hallucinations from **13.0% to 3.0%**.
- To support faculty and student research, delivered a **RAG research assistant** (**LlamaIndex, pgvector, FastAPI**) used by **40+ researchers**, raising **recall@10 from 61% to 88%**.
- To compare LLM vendors on equal terms, evaluated **GPT-4, Claude,** and **Gemini** through **LangChain**, selecting configurations that raised accuracy by **16%** and cut hallucinations by **19%**.
- To make AI work repeatable, wrote **documentation and reproducibility standards** adopted by **two follow-on LLM studies**, cutting onboarding time for new researchers to **under 2 days**.
- To turn pilots into supportable services, set up **MLflow**, **Docker**, and **CI/CD**, cutting release cycles from **2 weeks to 5 days** with **A/B testing**.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- To reduce member support load, delivered an enterprise **Azure OpenAI** assistant with **LangChain** and **Pinecone** RAG for **15K+ daily queries**, cutting resolution time by **38%** and raising ticket deflection by **25%**.
- To keep AI services reliable after launch, set up **monitoring**, **drift detection**, and automated retraining, sustaining **over 95% accuracy** for **12 months** across **200+ end users**.
- To integrate AI with clinical systems, containerized model services with **Docker, Kubernetes,** and **Jenkins CI/CD** behind **REST APIs**, improving uptime by **20%** and enabling real-time EHR integration.
- To automate clinical text review, built **BERT/BioBERT** extraction models on unstructured medical notes, reaching **80% accuracy** and cutting processing time by **55%**.
- To automate data workflows under **HIPAA**, designed **20+ pipelines** with **Apache Airflow** and **AWS**, raising throughput by **70%** and reaching **99.9% reliability**.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- To control access to sensitive health data, built **RESTful services** with **OAuth 2.0** authentication and **API versioning**, giving internal teams secure, consistent integrations.
- To align data work with business goals, partnered with stakeholders on the **Seven Plus Locations** project to map pipelines to KPIs, improving claims data accuracy.
- To reduce manual claims reconciliation, redesigned **claims-balancing workflows** in **Python** and **SQL**, cutting operational overhead by **60%**.
- To eliminate costly invoice errors, automated **medical invoice processing** with rule-based classification, reducing manual errors by **70%**.

PROJECTS

Agentic AI Ticket Automation System | SecureGPT, LLaMA, AWS SQS, JIRA, NIST-CSF

- To cut security-ticket triage time, built an **agentic AI** workflow with **SecureGPT** and **LLaMA** that auto-creates **JIRA** tickets under **NIST-CSF**, reaching **93% automation accuracy** and **70%** faster triage.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." *ACM WSDM '26*. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." *IEEE PST 2026*.